

Probability & Statistics 2

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Chapters

<u>Chapter 1:</u>	Some Basic Concepts
<u>Chapter 2:</u>	Fundamental Sampling Distributions and Data Descriptions
<u>Chapter 3:</u>	Confidence Intervals and Sample Size
<u>Chapter 4:</u>	Test of Hypothesis
<u>Chapter 5:</u>	Correlation and Regression





01

Some Basic Concepts



Statistics:

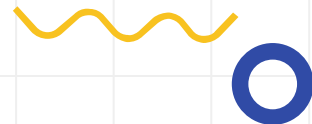
Is the science of :

collecting,
organizing,
presenting,
analyzing, and
interpreting data

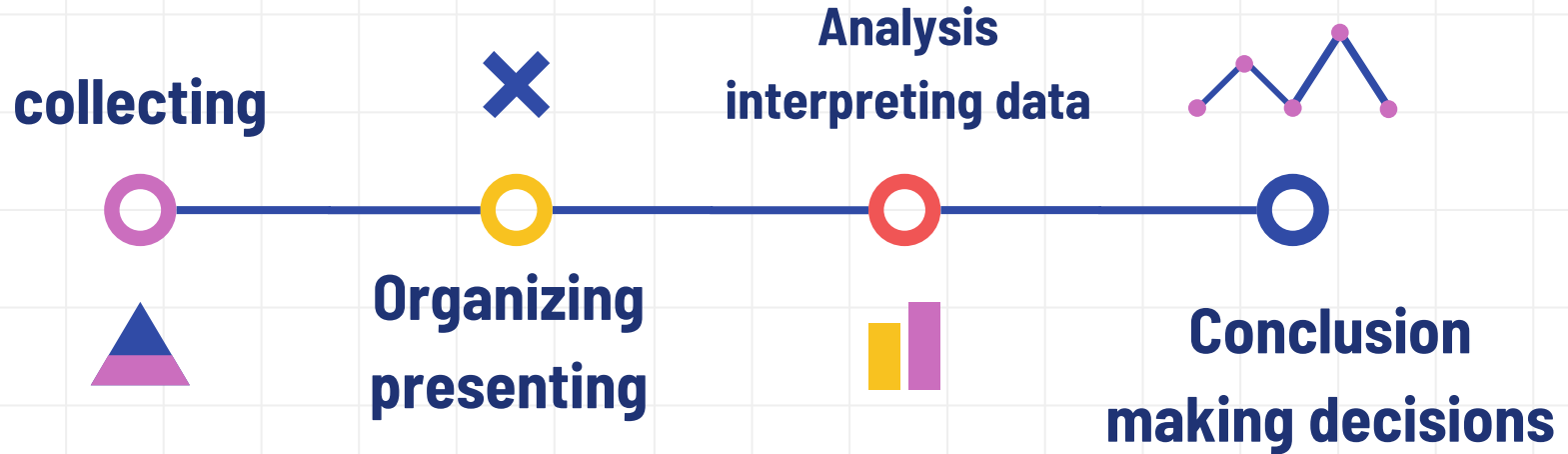
to assist in **making decisions** and
drawing conclusions.

Biostatistics:

If the information is obtained from
biological and medical sciences, then
we use the term biostatistics.



Statistical process



Why Study Statistics?

No matter what line of work you select, **you will find yourself faced with decisions where an understanding of data analysis is helpful.**

In order to make an informed decision, **you will need to be able to:**

1. **Determine** whether the existing information is sufficient or additional information is required.
2. **Gather additional information**, if it is needed, in such away that it does not provide misleading results.
3. **Summarize** the information in a useful informative manner.
4. **Analyze** the available information.
5. **Draw conclusions** and make inferences while assessing the risk of an incorrect conclusion.



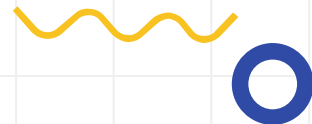
Statistics is the language of science and data.

Data:

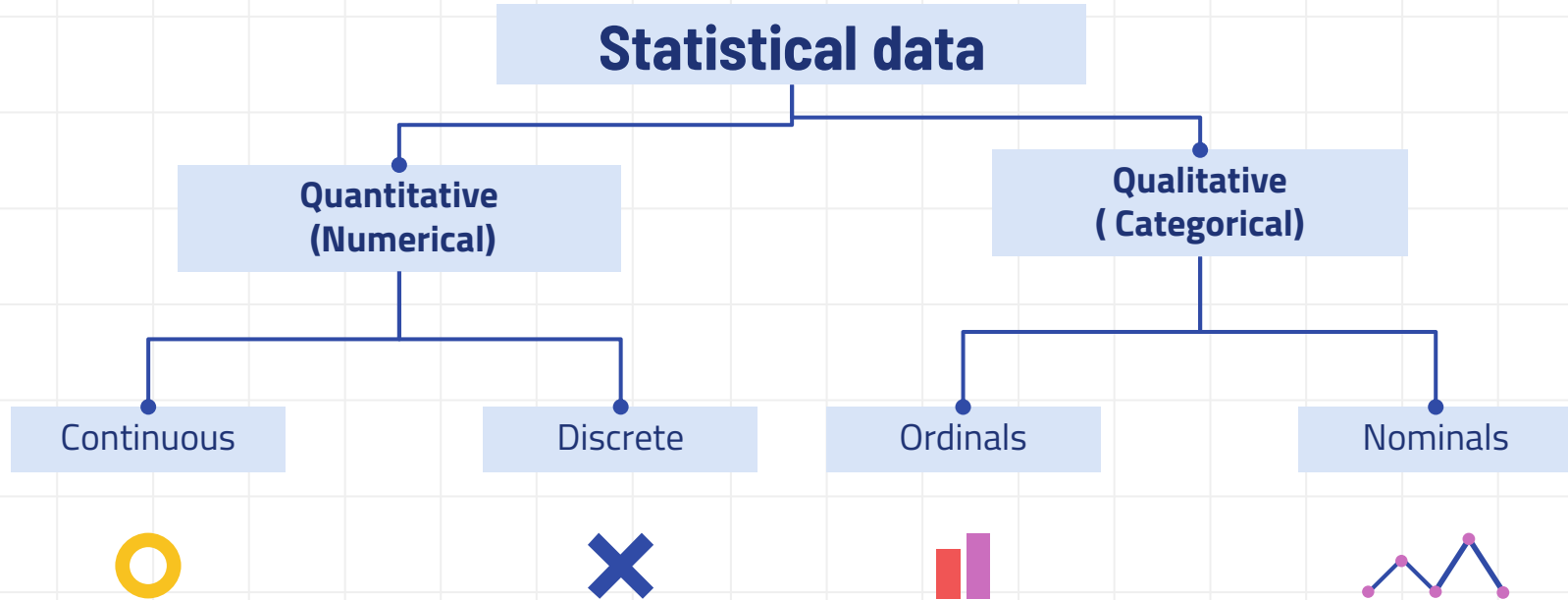
The raw material of Statistics is data. Data can be defined as a collection of facts or information from which conclusions may be drawn.

For example:

- When a botanist counts the number of Petals on a flower (counting).
- When a Zoologist weights a cow (measurement). They're simple



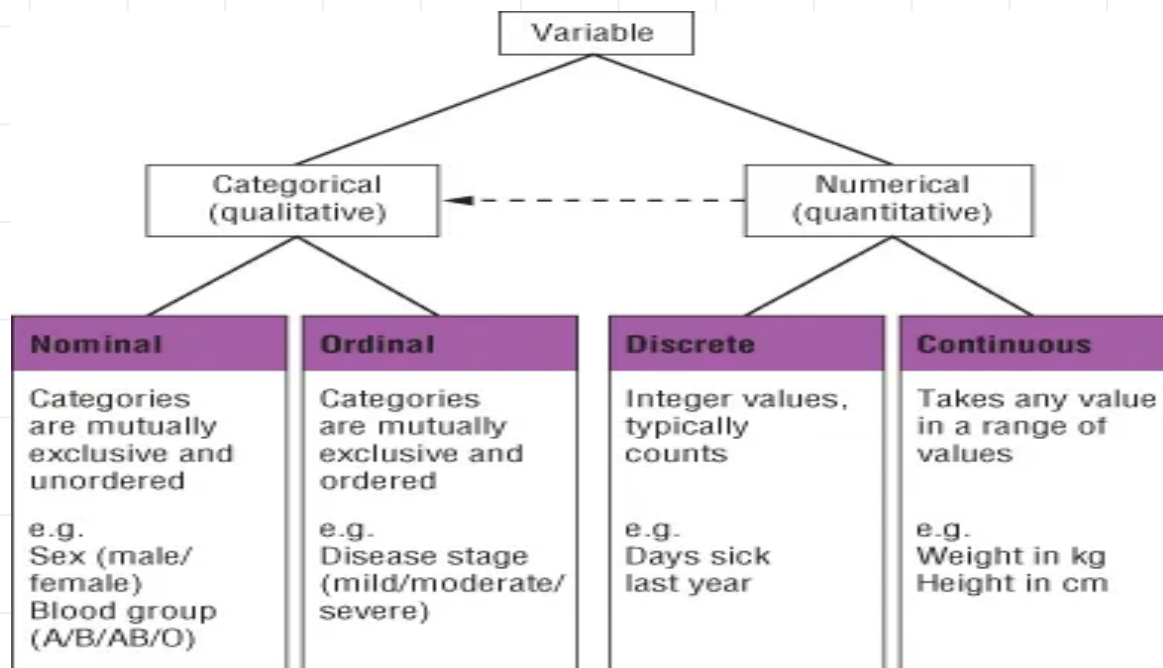
Classification of statistical data



Variable:

it is a characteristic that takes on different values in different persons, places, or things.

For example: heart rate, the heights of adult males, the weights of insects, the ages of trees seen in a forest.



1) Quantitative Variables:

A quantitative variable is a characteristic that can be measured. The values of a quantitative variable are numbers indicating **how much or how many of something**.

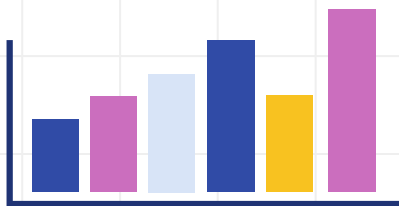
Examples:

- (i) Family Size (ii) No. of patients (iii) Weight (iv) height

(a) Discrete Variables:

There are jumps or gaps between the values.

- Examples: - Family size ($x = 1, 2, 3, \dots$)
- Number of patients ($x = 0, 1, 2, 3, \dots$)

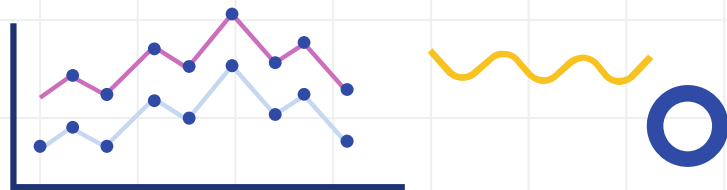


(b) Continuous Variables:

There are no gaps between the values.

A continuous variable can have any value within a certain interval of values.

- Examples: - Height ($140 < x < 190$)
- Blood sugar level ($10 < x < 15$)



2) Qualitative Variables:

The values of a qualitative variable are words or attributes indicating to which category an element belong.

Examples: - Blood type, Nationality , Students Grades ,Educational level

(a) Nominal Qualitative Variables:

A nominal variable classifies the observations into various mutually exclusive and collectively non-ranked categories.

Examples: Blood type (O, AB, A, B)
Nationality (Saudi, Egyptian, British, ...)
Sex (male, female)

(b) Ordinal Qualitative Variables:

An ordinal variable classifies the observations into various mutually exclusive and collectively ranked categories. The values of an ordinal variable are **categories that can be ordered**

Examples: Educational level (elementary, intermediate, ...)
Students grade (A, B, C, D, F)

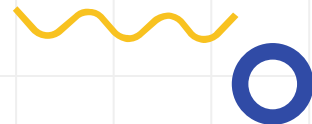


Types of statistics

Descriptive Statistics: are the methods of collecting, organizing, summarizing and measuring the collected data.

Inferential statistics:

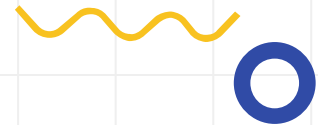
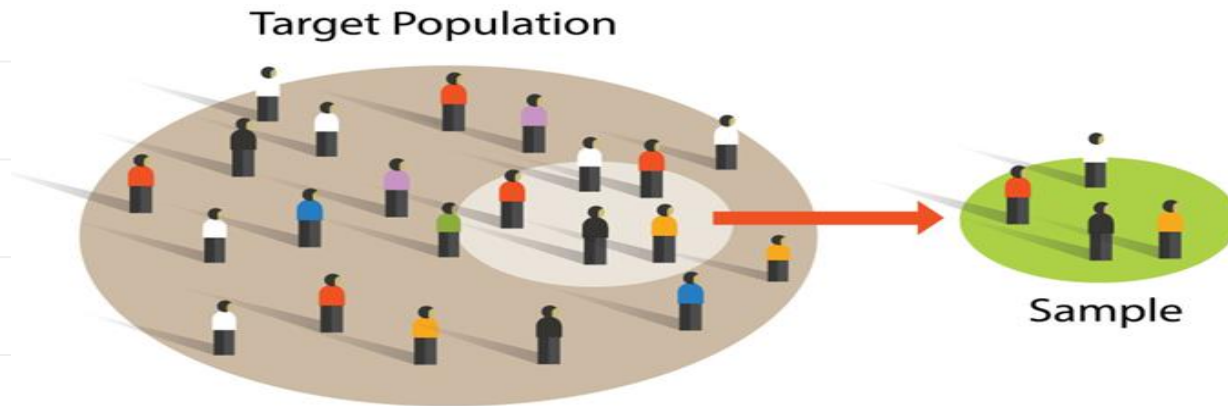
- methods which gives information about a population based on a sample,
- that is the process of using data obtained from a sample to make estimates or test hypotheses about the characteristics (parameter) of a population to all the production of the factory, but it is enough to **check a sample** of it on condition that this sample must represent the production of this factory completely, then we can generalize the results for all the production.

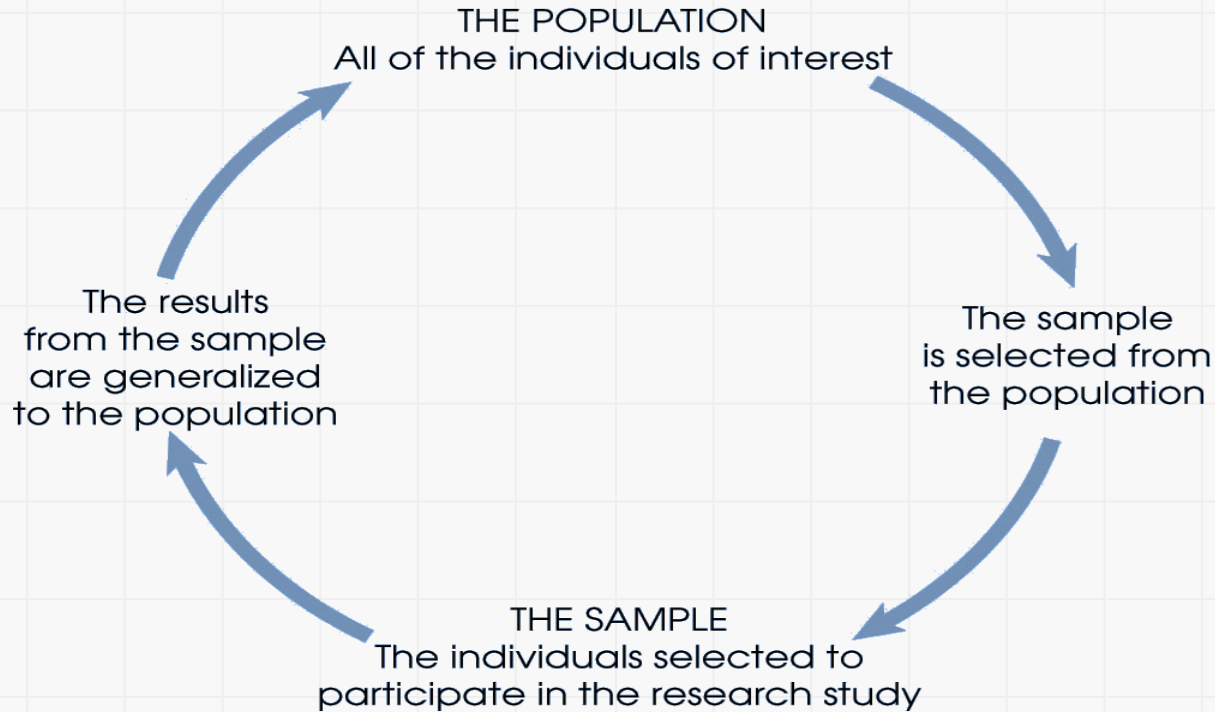


Populations and Samples

Population: It is the collection of all possible objects or measurements of interest.

Sample: Is a part of the population of interest





Data Collection and Sampling Techniques

Data can be collected in a variety of ways. **One of the most common** methods is through the **use of surveys**. Surveys can be done by using a variety of methods.

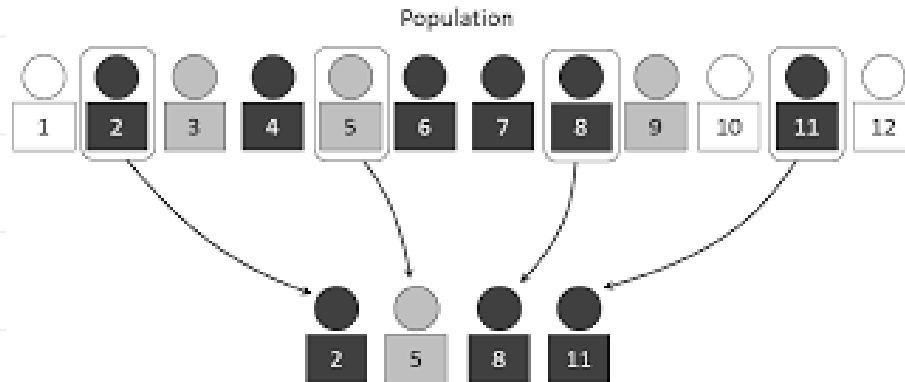
Three of the most common methods are

the telephone survey, the mailed questionnaire, and the personal interview.



Researchers use samples to collect data and information about a particular variable from a large population. Using samples saves time and money and in some cases enables the researcher to get more detailed information about a particular subject.

Random sample: is the sample whose elements are selected from the elements of a population. In this sample, each element must get **the same chance** of selecting. One such method is to number each subject in the population. Then place numbered cards in a bowl, mix them thoroughly, and select as many cards as needed. The subjects whose numbers are selected constitute the sample.

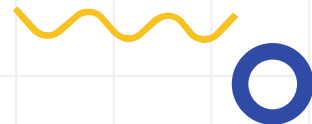


Stratified Sampling

Definition: Divides the population into groups (strata) based on important characteristics, then samples from each group.

Method: Randomly select samples within each stratum.

Example: A college president surveys first-year and second-year students separately to compare opinions.



Cluster Sampling

Definition: Divides the population into clusters (e.g., geographic areas or schools), then randomly selects entire clusters.

Method: All members of selected clusters are included in the sample.

Example: A researcher randomly selects 2 out of 10 apartment buildings and surveys all residents in those buildings.

Differences

Stratified Sampling: Ensures representation from each group.

Cluster Sampling: Focuses on entire clusters, often used for large populations or geographic areas.



1. What is the primary purpose of statistics?

- A) To collect data
- B) To assist in making decisions and drawing conclusions
- C) To present data in graphs
- D) To memorize facts

Answer: B

2. What term is used for data obtained from biological and medical sciences?

- A) Inferential Statistics
- B) Descriptive Statistics
- C) Biostatistics
- D) Demographics

Answer: C



3. Which of the following is an example of discrete data?

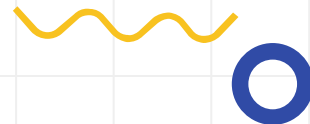
- A) Height of students
- B) Number of patients in a hospital
- C) Weight of apples
- D) Blood sugar levels

Answer: B

4. What is stratified sampling?

- A) Sampling from a random selection of clusters
- B) Dividing the population into strata and sampling from each
- C) Collecting data from the entire population
- D) Using only one group for sampling

Answer: B



True or False

1. Statistics is only concerned with collecting data.

Answer: False

2. Qualitative data can be number.

Answer: False

3. Cluster sampling involves selecting entire groups rather than individual subjects.

Answer: True

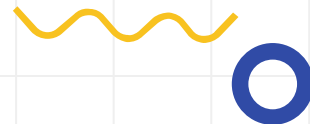
4. A population includes all possible measurements of interest.

Answer: True

5. Inferential statistics is used to summarize data.

Answer: False

(This is the role of descriptive statistics.)

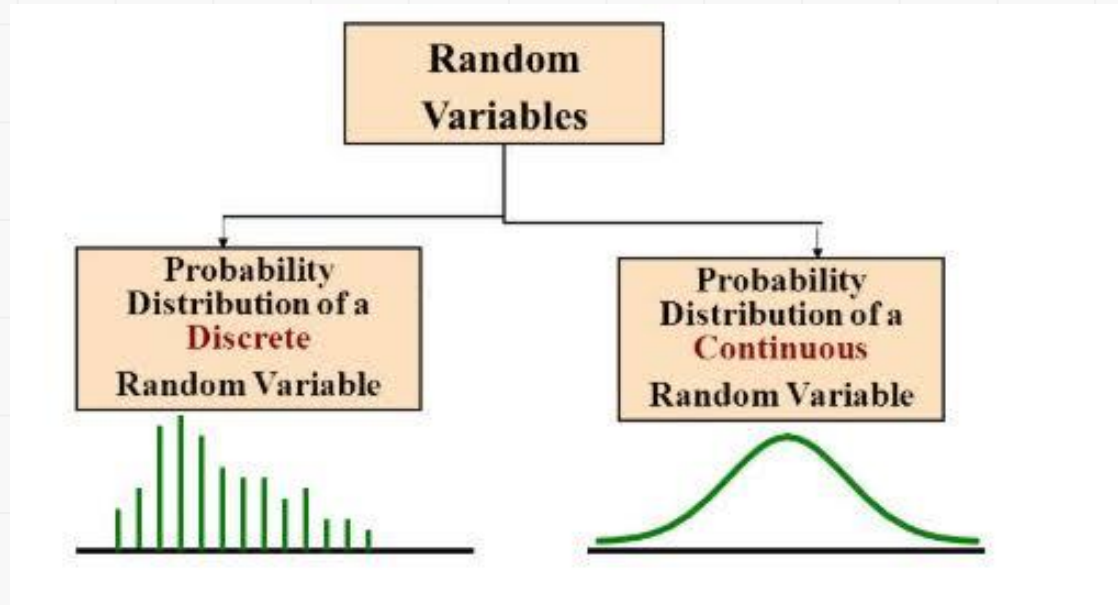


Random Variables

- **A random variable** is a variable whose value is a numerical outcome of a random phenomenon.
- **For example:** Flip three coins and let (X) represent the number of heads. (X) is a random variable.
- **The sample space** (S) lists the possible values of the random variable (X)



Variables whose values are determined by chance are called **random variables**.



Discrete Random Variables

Probability Density Function (pdf) :

For a random variable X of the discrete type, the probability $P(X = x)$ is frequently denoted by $f(x)$, and this function $f(x)$ is called probability density (or mass) function.

$$f(x) = P(X = x)$$



The pdf $f(x)$ of a discrete random variable X is a function that satisfies the following properties:

$$1) 0 \leq f(x) \leq 1, \quad x \in R.$$

$$2) \sum_x f(x) = 1$$

$$3) P(X \in A) = \sum_{x \in A} f(x) \quad \text{where } A \subset R.$$



Cumulative distribution function (Cdf) :

Is the probability that the discrete r.v X will take a value less than or equal to x

$$F(x) = P(X \leq x) = \sum_{\leq x} f(x)$$



The mean, variance and standard deviation of a discrete r.v X can be obtain respectively by:

$$\mu = E[X] = \sum_{all\ x} xf(x)$$

is the **arithmetic average** or the **mean** of X.

$$\begin{aligned}\sigma^2 = V[X] &= E[(X - \mu)^2] = \sum_{all\ x} (X - \mu)^2 f(x) \\ &= E[X^2] - \mu^2 = [\sum_{all\ x} x^2 f(x)] - \mu^2\end{aligned}$$

The variance is a measure of **the dispersion or spread** for the random variable X.

The positive square root of the variance is called the **standard deviation** of X and it is denoted by

$$\sigma = \sqrt{\sigma^2}$$



Remember:

$$(1) E[aX \pm b] = aE[X] \pm b$$

$$(2) V[aX \pm b] = a^2 V[X], \text{ where } a, b \text{ are constants}$$



Example

Let X have the pdf

$$f(x) = \frac{x}{10}, \quad x = 1, 2, 3, 4.$$

Find:

(i) $E[X]$

(ii) $E[X^2]$

(iii) $E[2X - 1]$

(iv) $E[X(5 - X)]$



Solution

$$\text{i) } \mu = E[X] = \sum_{x=1}^{x=4} x \left(\frac{x}{10} \right) = 1 \left(\frac{1}{10} \right) + 2 \left(\frac{2}{10} \right) + 3 \left(\frac{3}{10} \right) + 4 \left(\frac{4}{10} \right) = 3$$

$$\text{ii) } E[X^2] = \sum_{x=1}^{x=4} x^2 \left(\frac{x}{10} \right) = 1^2 \left(\frac{1}{10} \right) + 2^2 \left(\frac{2}{10} \right) + 3^2 \left(\frac{3}{10} \right) + 4^2 \left(\frac{4}{10} \right) = 10$$

$$\text{iii) } E[2X - 1] = 2E[X] - 1 = 2(3) - 1 = 5$$

$$\text{(iv) } E[X(5 - X)] = E[5X - X^2] = 5E[X] - E[X^2] = 5(3) - 10 = 5$$



Example

Calculate the mean and the standard deviation of X if:

Let X be the number on each ball. The probability distribution is

Number on ball X	3	4	5
Probability $P(X)$	$\frac{2}{3}$	$\frac{1}{3}$	$\frac{2}{3}$



Solution

The mean, variance, and standard deviation can also be found by using vertical columns, as shown.

X	$P(X)$	$X \cdot P(X)$	$X^2 \cdot P(X)$
3	0.4	1.2	3.6
4	0.2	0.8	3.2
5	0.4	2.0	10
		$\Sigma X \cdot P(X) = 4.0$	16.8

Find the mean by summing the $\Sigma X \cdot P(X)$ column, and find the variance by summing the $X^2 \cdot P(X)$ column and subtracting the square of the mean.

$$\sigma^2 = 16.8 - 4^2 = 16.8 - 16 = 0.8$$

and

$$\sigma = \sqrt{0.8} = 0.894$$

Continuous Random Variables

Probability Density Function (pdf) :

$f(x)$ is a probability density function (pdf) of a random variable X of the continuous type, with space R that is an interval or union of intervals, if the following three conditions are satisfied:

1) $0 \leq f(x) \leq 1, \quad \forall x \in R$

2) $\int_R f(x)dx = 1.$

3) The probability of the event $(X \in A)$ is

$$P(X \in A) = \int_A f(x)dx$$



The cumulative distribution function (cdf) of a random variable X of the continuous type is given by

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(x)dx$$

$$\text{Note: } P(a < X \leq b) = P(a \leq X \leq b) = P(a < X < b) = \int_a^b f(x)dx$$

$$= F(b) - F(a)$$



The mean, variance and standard deviation of a continuous r.v X can be obtained respectively by:

$$\mu = E[X] = \int_{-\infty}^{\infty} xf(x)dx$$

$$\begin{aligned}\sigma^2 &= V[X] = E[X^2] - \mu^2 = \left[\int_{-\infty}^{\infty} x^2 f(x)dx \right] - \mu^2 \\ &= E[(X - \mu)^2] = \int_{-\infty}^{\infty} (x - \mu)^2 f(x)dx\end{aligned}$$

$$\sigma = \sqrt{\sigma^2}$$



Example

Let X be a continuous r. v. such that

$$f(x) = \begin{cases} \frac{x}{2} & 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

Find μ and σ .



Solution

$$\mu = E[X] = \int_0^2 xf(x)dx = \int_0^2 \frac{x^2}{2} dx = \frac{4}{3}$$

$$E[X^2] = \int_0^2 x^2 f(x)dx = \int_0^2 \frac{x^3}{2} dx = 2$$

$$\therefore \sigma^2 = E[X^2] - \mu^2 = 2 - \left(\frac{4}{3}\right)^2 = \frac{2}{9} \quad \rightarrow \quad \sigma = \sqrt{\frac{2}{9}}$$



Example

Calculate the mean and the variance of Y if:

$$f(y) = \frac{3}{8}Y^2, 0 \leq y \leq 2$$



Example

$$\begin{aligned} E(Y) &= \int_{-\infty}^{\infty} y f(y) dy \\ &= \int_0^2 y \left(\frac{3}{8} \right) y^2 dy \\ &= \left(\frac{3}{8} \right) \left(\frac{1}{4} \right) y^4 \Big|_0^2 = 1.5. \end{aligned}$$



Solution

The variance of Y can be found once we determine $E(Y^2)$. In this case,

$$\begin{aligned} E(Y^2) &= \int_{-\infty}^{\infty} y^2 f(y) dy \\ &= \int_0^2 y^2 \left(\frac{3}{8}\right) y^2 dy \\ &= \left(\frac{3}{8}\right) \left(\frac{1}{5}\right) y^5 \Big|_0^2 = 2.4. \end{aligned}$$

Thus, $\sigma^2 = V(Y) = E(Y^2) - [E(Y)]^2 = 2.4 - (1.5)^2 = 0.15$.





Thanks!

